Week 01: What Is Artificial Intelligence? Definitions, Aims and Measures

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Before asking whether machines can think, it helps to ask what artificial intelligence is supposed to be. This week compares definitions of AI, separates strong from weak AI and examines formal attempts to measure intelligence, which make the hidden assumptions of informal definitions explicit [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to distinguish the four approaches in the classification of Russell and Norvig. They should be able to explain the difference between strong and weak AI, and between both and general AI. They should also be able to compute a simplicity-weighted intelligence score in the spirit of Legg and Hutter. Finally, they should be able to discuss Chollet's view of intelligence as skill-acquisition efficiency.

Lecture

A field defined by an ambition

The proposal for the 1955 Dartmouth summer project gave the field its name. It rested on the conjecture that every aspect of learning and intelligence can in principle be described precisely enough for a machine to simulate it [4]. The conjecture is philosophical as much as technical: It assumes that intelligence is a matter of describable processes rather than of a particular biological substrate.

Russell and Norvig classify definitions of AI along two dimensions [1]. The first asks whether a definition concerns thought or behaviour. The second asks whether success is measured against human performance or against an ideal of rationality. The result is four approaches. Thinking humanly is the programme of cognitive modelling, which compares programs with human thought processes. Thinking rationally follows the laws of thought tradition, in which intelligence is correct inference. Acting humanly is the approach of the Turing test, which Week

2 examines. Acting rationally, the approach that Russell and Norvig adopt, defines an intelligent agent as one that acts to achieve the best expected outcome. Figure 1.1 summarises the classification.

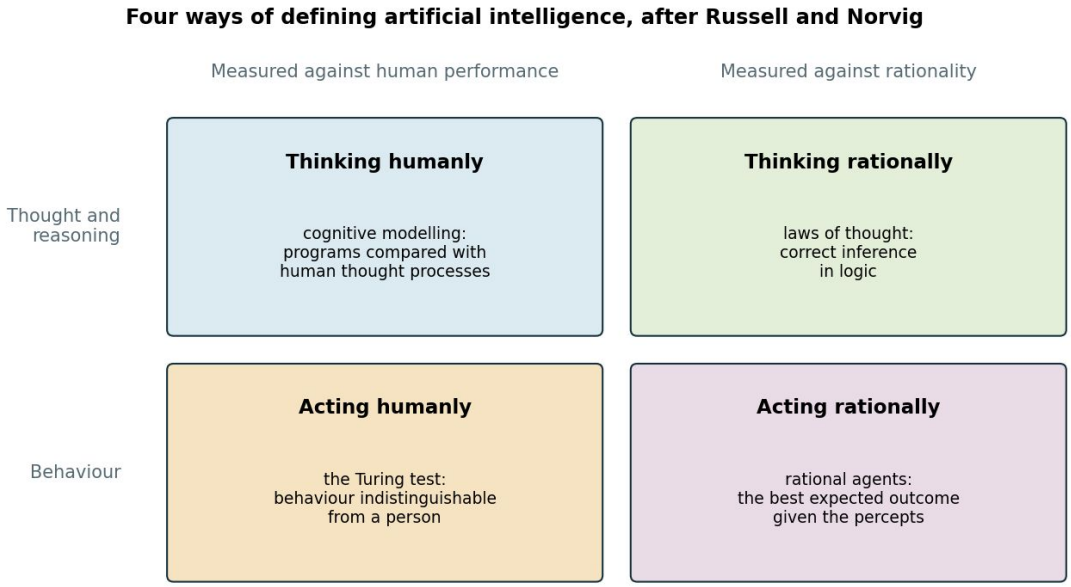


Figure 1.1. The classification of definitions of AI by Russell and Norvig along two dimensions: thought versus behaviour and human performance versus rationality [1].

Strong and weak AI

Searle introduced a distinction that frames much of this course [2]. According to weak AI, the computer is a powerful tool for studying the mind, for example by testing hypotheses about cognition with simulations. According to strong AI, an appropriately programmed computer does not merely simulate a mind: It literally has cognitive states and understands. Searle accepted weak AI and rejected strong AI, and his argument is the topic of Week 4.

The distinction is often confused with a different one. Narrow AI performs specific tasks, while general AI would perform well across a wide range of tasks. This is a distinction of capability, not of metaphysics. A system could be general without having a mind in Searle's sense, and a narrow system could in principle be a candidate for strong AI. Keeping the two axes apart prevents a common error, namely inferring conclusions about minds from improvements in performance.

The philosophy of AI therefore asks several kinds of question. Conceptual questions ask what thinking, understanding and intelligence are. Metaphysical questions ask whether a machine could have a mind. Epistemological questions ask how anyone could know. Methodological questions ask what AI research shows about minds in general.

Measuring intelligence

Informal definitions leave room for disagreement, so some researchers have tried to formalise intelligence. Legg and Hutter collected many informal definitions and proposed universal intelligence [3]. It is the expected performance of an agent across all computable environments. Each environment is weighted by two to the power of minus its Kolmogorov complexity. Simple environments therefore count more than complex ones. The measure cannot be computed exactly, because Kolmogorov complexity is not computable, but it can be approximated. Its structure exposes a choice that informal definitions hide: which environments matter, and how much.

Chollet argued that performance on any fixed set of tasks measures skill, which can be bought with prior knowledge or unlimited training data [5]. He proposed to define intelligence as skill-acquisition efficiency: how efficiently a system turns its priors and experience into skill on tasks that require generalisation. His benchmark of abstract reasoning puzzles, ARC, was designed on this principle. The interactive lab of this week shows how the ranking of four agents changes when the weighting of environments changes. It makes visible how any intelligence score depends on its assumptions.

Why definitions matter

Definitions are not neutral. A benchmark score that improves on narrow tasks may be reported as progress towards general intelligence, and a behavioural definition may be read as settling questions about understanding. Each later week of the course tests one of these inferences. Weeks 2 to 4 examine behaviour, computation and understanding. Weeks 5 to 8 turn to meaning, cognitive architecture, embodiment and common sense. Week 9 examines arguments that machines face principled limits. Weeks 10 and 11 address consciousness. Weeks 12 to 14 apply the tools of the course to understanding, creativity and intentionality in present systems.

> ***Pause and reflect.*** *Which of the four approaches best describes the AI systems in your own research? Would your answer change if the systems became much more capable?*

Hands-on components

Interactive lab: Part A asks for the approach behind eight descriptions of AI systems. Part B computes a simplicity-weighted intelligence score for four agents in six environments, in the spirit of Legg and Hutter. Change the strength of the simplicity prior and watch the ranking change [1, 3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-01/lab.html

Colab notebook: It builds six sequence-prediction environments from short programs and approximates their description length by compression. It runs four prediction agents in each environment and computes simplicity-weighted scores, following the structure of the universal

intelligence measure [3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-01/NB01_what_is_ai.ipynb

Reflection prompts

1. Write your own definition of intelligence in one sentence. Which of the four approaches does it belong to, and what would count as evidence against it?
2. Is it possible for a system to be general but not strong in Searle's sense? Give an example or an argument.
3. Which environments would you include in an intelligence measure for systems in your field, and how would you weight them?

Weekly task and submission

What to do. Write a short essay that compares two definitions of intelligence from this week. A natural pair is the universal intelligence measure of Legg and Hutter and Chollet's view of intelligence as skill-acquisition efficiency. First, state each definition in your own words. Second, name one assumption that each definition makes explicit or hides. Third, apply both definitions to one AI system of your choice and say whether they judge it in the same way.

Where to look. Lecture: sections "Measuring intelligence" and "Why definitions matter". Interactive lab: Part B, where the strength of the simplicity prior changes the ranking of four agents. Colab notebook: Section 3 and Exercise 2, which show the same effect in Python.

Example. A chess engine plays one game very well but cannot learn a new game on its own. Under the universal intelligence measure, it scores low because it fails in most environments. Under Chollet's view, it scores low because it needed vast experience and built-in knowledge to acquire one skill. Both definitions give a low score, but for different reasons.

What to submit. About 500 words that cite at least three works in square brackets, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.

[2] Searle, J. R. (1980). Minds, brains, and programs. Behavioral and Brain Sciences, 3(3), 417-424. https://doi.org/10.1017/S0140525X00005756

[3] Legg, S., & Hutter, M. (2007). Universal intelligence: A definition of machine intelligence. Minds and Machines, 17(4), 391-444. https://doi.org/10.1007/s11023-007-9079-x

[4] McCarthy, J., Minsky, M. L., Rochester, N., & Shannon, C. E. (2006). A proposal for the Dartmouth summer research project on artificial intelligence, August 31, 1955. AI Magazine, 27(4), 12-14. https://doi.org/10.1609/aimag.v27i4.1904

[5] Chollet, F. (2019). On the measure of intelligence. arXiv preprint arXiv:1911.01547. https://arxiv.org/abs/1911.01547